

Experiment:2 - Monoalphabetic Substitution Cipher

```
#include <stdio.h>

int main()
{
    char text[100];
    char key[27];
    int i, pos;
    printf("Enter plaintext: ");
    fgets(text, sizeof(text), stdin);
    printf("Enter 26-letter substitution key: ");
    scanf("%26s", key);
    for (i = 0; text[i] != '\0'; i++)
    {
        if (text[i] >= 'A' && text[i] <= 'Z')
        {
            pos = text[i] - 'A';
            text[i] = key[pos];
        }
        else if (text[i] >= 'a' && text[i] <= 'z')
        {
            pos = text[i] - 'a';
            text[i] = key[pos];
        }
    }
    printf("Ciphertext: %s", text);

    return 0;}
```



C:\Users\ebina\OneDrive\Des



Enter plaintext: Hello

Enter 26-letter substitution key: QWERTYUIOPASDFGHJKLZXCVBNM

Ciphertext: ITSSG

Process exited after 11.81 seconds with return value 0

Press any key to continue . . .